

Ahmet Taha Albayrak

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PERSONAL SUMMARY

Innovative AI Engineer and Co-Founder with over 8 years of professional experience in architecting and deploying high-impact artificial intelligence solutions. Expert in **Computer Vision**, **Multimodal Generative AI**, and **Agentic AI**, with a specialized focus on building billion-scale ETL pipelines and autonomous agentic workflows. Proven track record in leading technical teams at NuveMind, Pavo Group and Tübitak, delivering scalable systems for biometrics, fraud detection, and OS-level AI integrations. Skilled in bridging the gap between cutting-edge research and production-ready ML practices.

EDUCATION

Middle East Technical University

MSc in Computer Engineering CGPA: 3.34/4.00

Ankara, Turkey

Sep 2021- Sep 2023(Leave)

Bilkent University

B.Sc. in Computer Engineering; CGPA: 3.32/4.00

Ankara, Turkey

Sep 2013 – Jun 2018

WORK EXPERIENCE

NuveMind Information Technologies

AI Engineer, Co-Founder

Ankara, Türkiye

Mar 2025 – Present

- Multimodal Generative AI:** Led the end-to-end architectural design, optimization, and commercialization of multimodal generative AI solutions by integrating advanced image processing and natural language technologies.
- Autonomous AI Agents:** Architected and deployed scalable AI agentic systems capable of autonomously executing complex workflows through seamless interactions with external tools and APIs.
- Team Coordination & Agile Development:** Coordinated software and AI teams during the integration of agent-based systems and multimodal models, establishing a sustainable development lifecycle.
- Multimodal Generative AI:** Led the end-to-end architectural design, optimization, and commercialization of multimodal generative AI solutions by integrating advanced image processing and natural language technologies.
- Autonomous AI Agents:** Architected and deployed scalable AI agentic systems capable of autonomously executing complex workflows through seamless interactions with external tools and APIs.
- Team Coordination & Agile Development:** Coordinated software and AI teams during the integration of agent-based systems and multimodal models, establishing a sustainable development lifecycle.

Pavo Group

Senior Machine Learning Engineer

Ankara, Türkiye

Nov 2021 – Jan 2025

- MLOps Architecture:** Implemented comprehensive MLOps practices across, reducing model deployment time by 60% and increasing model iteration speed by 40%.
- Edge Inference Optimization:** Optimized real-time machine learning algorithms, achieving inference speeds of 30 fps on edge devices, a 3x improvement over previous implementations.
- Multi-Task Learning (MTL):** Developed and implemented MTL models for 2D/3D pose estimation and object detection achieving a 30% improvement in accuracy and 40% reduction in computational cost.
- IR Sensor Integration:** Integrated IR camera technology into machine learning pipelines, enhancing data collection by 50% and improving performance for low-light tasks.
- Large-Scale Data Processing:** Conducted data analysis and preprocessing on datasets of over 1 million images leading to a 35% increase in model accuracy.

TÜBİTAK SPACE RESEARCH INSTITUTE

Researcher

Ankara, Türkiye

Jul 2020 – Nov 2021

- Distributed Computing:** Explored the use of distributed volunteer computing to enhance system capabilities and published a paper on hyperparameter optimization.
- Microservices & Containerization:** Utilized Docker to containerize and deploy services, simplifying the deployment process and improving scalability.
- Algorithm Development:** Worked on developing machine learning algorithms using Python, PyTorch, and MXNet for complex research tasks.

GCLab Engineering

Computer Engineer

Ankara, Türkiye

Oct 2018 – Jul 2020

- **Computer Vision & Image Processing:** Worked within the TÜBİTAK UZAY Image Processing team, focusing on the development of vision-based systems and algorithmic research.
- **Semantic Search & Indexing:** Implemented semantic search capabilities for images and sound, improving the indexing process to be faster and more compact.
- **Multimodal User Queries:** Developed systems for supporting multimodal user queries and improved internet-facing search capabilities.

AWARDS & ACHIEVEMENTS

Bilkent University 2017-2018 Senior Project Data Science Award

National University Admission Exam (YKS): Ranked 753th in Mathematics and Science among ca. 1.8 million candidates

Enderun Science High School Salutatorian Award: Graduated as the highest ranked student in my class.

PUBLICATIONS

A.T. Albayrak, İ. Atıl, "Hyper Parameter Optimization for Deep Learning using Volunteer Computing" ,6th High Performance Computing Conference, Oct. 8-9, 2020.

SKILLS

Programming: C, C++, Python, MATLAB, SQL

Technologies:Git, Arduino, Simulink, Tensorflow, PyTorch, Numpy, CUDA

Languages: Turkish (Native), English (Professional)